

Financial Time Series

46-929

Homework #4

Due: Thursday, April 22, 2021, 1:30pm.

1. (Forecasting stock prices)

Download S&P 500 index data for 1/1/2020 to 4/16/2021. Using log-returns check that the time series is weakly stationary. Using data for 1/1/2020 to 4/1/2021 fit an ARIMA model and forecast the index from 4/2/2021 to 4/16/2021.

2. (Simulating ARCH(1))

Consider an ARCH(1) model

$$a_t = \epsilon_t \sigma_t,$$
$$\sigma_t^2 = 1 + \alpha a_{t-1}^2,$$

where $\{\epsilon_t\}$ are i.i.d. $N(0,1)$ random variables. Simulate and plot $\{a_t^2, 1 \leq t \leq 1000\}$ assuming $\alpha = .2, .4, .6$. How, if at all, do the plots differ? Do the plots change if the $N(0,1)$ distribution is changed to a $t_4/\sqrt{2}$ distribution? (Note, the t_4 distribution has variance 2).

3. (AR(1) + ARCH(1) Process)

Suppose that ϵ_t is an iid $WN(0, 1)$ process, that

$$a_t = \epsilon_t \sqrt{1.5 + 0.30a_{t-1}^2},$$

and that

$$Y_t = 1 + 0.6Y_{t-1} + a_t.$$

- a) Find the mean of Y_t .
- b) Find the variance of Y_t .
- c) Find the autocorrelation function of Y_t .
- d) Find the autocorrelation function of a_t .
- e) Find the autocorrelation function of a_t^2 .

4. (GARCH Modeling)

Using the S&P 500 index data from problem 1, fit a constant mean μ and GARCH volatility model using the ARCH python package. Note the coefficients, their standard error and the p-values for testing if any of them are 0.

Refer to the documents and examples for the ARCH package at <https://arch.readthedocs.io/en/latest/univariate/introduction.html>. This type of problem is covered in the examples.